Pengetahuan QA/QC

Definisi QA/QC

Quality assurance Merupakan activitas yang direncanakan dan sistematis (teratur menurut system) yang diimplementasikan dalam suatu quality system sehingga kebutuhan quality untuk suatu product atau jasa akan terpenuhi atau terjamin.

Quality control Merupakan teknik pengawasan dan activitas yang digunakan untuk memenuhi persyaratan mutu.

Perbedaan QA/QC

Quality Assurance (penjaminan mutu) adalah semua tindakan terencana, sistematis dan didemonstrasikan untuk meyakinkan pelanggan bahwa persyaratan yang ditetapkan "akan dijamin" tercapai. Salah satu elemen dari QA adalah QC. Elemen yang lain yaitu: Planning, organization for quality, Established Procedure, Supplier Selection, Corrective Action, Document control, training.

Quality Control (pengendalian mutu) adalah kegiatan untuk memantau, mengevaluasi dan menindaklanjuti agar persyaratan mutu yang ditetapkan tercapai (Product, process, service, inspection, testing, sampling, measurement dan calibration).

Jadi intinya adalah
QA : Conceptor (Penjamin Mutu)
QC : Executor/Operator (Pengendali Mutu)

Tugas QA/QC

QA (Quality Assurance): Tugasnya memahami specification customer dan standard yang berhubungan dengan produk, kemudian membuat / menentukan cara inspectionnya (berupa prosedur) dan mendokumentasi hasil inspectionnya (manufacturing data report). QA lebih banyak paper work, umumnya memiliki skill inspection yang baik dan skill menulis procedure dan familiar dengan engineering & industrial standards.

QC (Quality Control): Tugasnya melakukan inspection berdasarkan prosedur yang dibuat dan disahkan oleh QA. QC lebih banyak melakukan inspection pada process manufacturing dan membuat laporannya.

Ada 4 langkah QA/QC program:

1. **Training (Pelatihan)**
2. **Pre-Planning of work (Perencanaan Kerja)**
3. **The Job (Pekerjaan)**
4. **Post Job Review (Meninjau ulang Pekerjaan):**

CODE : adalah Dokumen yang mengatur persyaratan – persyaratan minimal dari suatu desain, material, fabrikasi, instalasi, pengetesan, inspeksi dalam sistem perpipaan.

STANDARD : adalah Dokumen yang mengandung peraturan desain, konstruksi dan persyaratan individu dari komponen perpipaan seperti pipa, elbow, fitting, flange, valve, gasket dll. Pemakaian Code dan Standard tersebut harus sesuai dengan proses pada sistem perpipaan yang digunakan.

SPESIFIKASI : Petunjuk teknis maupun petunjuk pelaksanaan dari suatu project berdasarkan keperluan product.

ASME American Society of Mechanical Engineers
- Section I Power Boilers
- Section II Material Specifications
  - Part A - Ferrous
  - Part B - Nonferrous
  - Part C - Welding Rods, Electrodes and Filler Metals
- Section V Nondestructive Examination
- Section VIII Unfired Pressure Vessels
  - Division 1 - Pressure Vessels
  - Division 2 - Alternative Rules Section IX
- Section IX Welding and Brazing Qualifications

ASME/ANSI
- B16.5 Pipe Flange & Flange Fitting
- B16.9 Wrought Steel Butt Welding Fittings
- B16.10 Valve Dimensions
- B16.11 Forged Fitting
- B16.20 Metalic Gasket for Pipe Flange
- B16.21 Nonmetalic Gasket
- B16.25 Butt-Welding Ends
- B16.34 Valve, Flange, Threaded, And Welding End
- B16.47 Large Diameter Steel Flanges
- B31.1 Power Piping
- B31.3 Chemical Plant and Petroleum Refinery Piping
- B31.8 Gas Transmission and Distribution Piping Systems
- B36.19 Stainless Steel Pipe

ANSI American National Standards Institute

API American Petroleum Institute
- 5L Specification for Line Pipe
- 510 Pressure Vessel Inspection Code
- 570 Piping Inspection code (Inspection, Repair, Alteration, and Rerating of In-
- 610 Centrifugal Pumps for General Refinery Service
- 686 Machinery Installation and Installation Design
- 650 Welded steel tank for oil storage
- 1104 Welding of Pipelines and Related Facilities

AWS American Welding Society
- A2.4 Symbols for Welding, Brazing and Non-destructive Examination
- A3.0 Standard Welding Terms and Definitions
- D1.1 Structural Welding Code Steel

ASNT American Society for Non-Destructive Testing
- TC-1A NDT Personnel Qualification and Certification

ASTM American Society for Testing and Materials
AWWA American Water Works Association
AISC American Institute of Steel Construction
CEMA Conveyor Equipment Manufacturers Association
MSS Manufacturers Standardization Society
NACE National Association of Corrosion Engineers
NFPA National Fire Protection Association
SSPC Steel Structures Painting Council

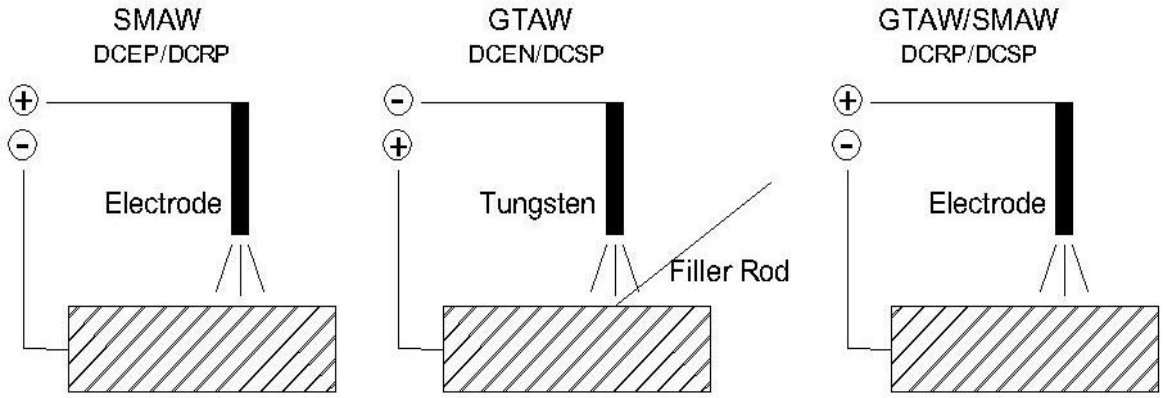
ISO International Standard Organization
- 9000 Quality management systems —Fondamentals & Vocabulary
- 9001 Quality management systems —Requirement

WELDING PROCESS

Welding Process ASME-IX		Welding ProcessEN288-3	
OFW	Qxyfuel Gas Welding	**111**	Meta-arc Welding; with covered electrode
SMAW	Shielded Metal-Arc Welding →Polarity (DCEP)	**114**	Flux-cored metal-arc; welding without gas shield
SAW	Submerged-Arc Welding	**121**	Submerged arc welding; with electrode
GMAW/MAG	Gas Metal-Arc Welding/Metal Active Gas	**122**	Submerged arc welding; with strip electrode
FCAW	Flux Cored-Arc Welding	**131**	Metal-arc inert gas welding (MIG welding)
GTAW/ MIG	Gas Tungsten-Arc Welding/Meteal Inert Gas	**135**	Metal-arc active gas welding (MAG welding)
PAW	Plasma-Arc Welding	**136**	Flux-cored wire metal-arc welding; with active gas shield
ESW	Electroslag Welding	**137**	Flux-cored wire metal-arc metal; with inert gas shield
EGW	Electrogas Welding	**141**	Tungsten inert gas welding (TIG welding)
EBW	Electron Beam Welding	**15**	Plasma arc welding
SW	Stud Welding		
ICDFW	Innertia and Continuous Drive Friction Welding		
RW	Resistance Welding		
LBW	Laser Beam Welding		
FW	Flash Welding		

POLARITY (masukan panas)

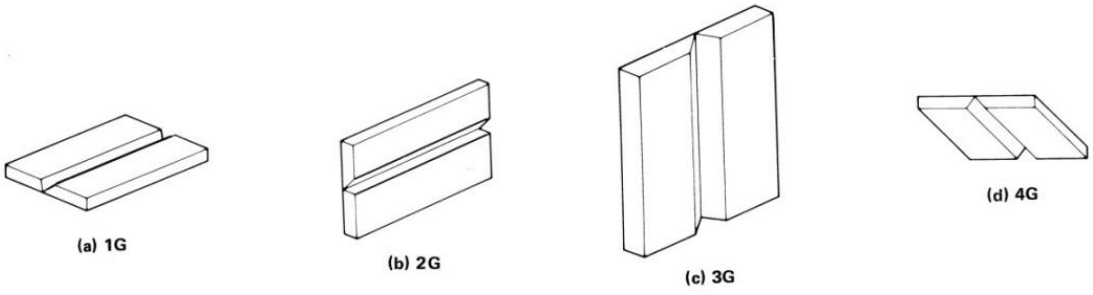
DCEP/DCRP = direct current electrode positive/ direct current reverse polarity
DCEN/DCSP = direct current electrode negative/ direct current straight polarity
AC = Alternating Current
DC = Direct Current



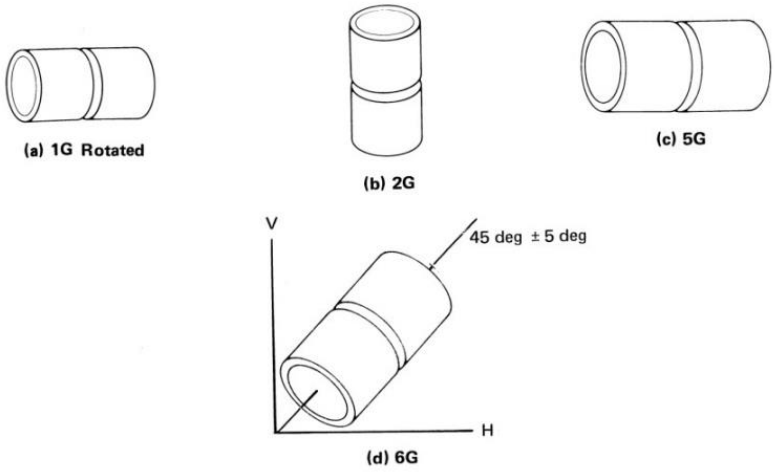
Process Meta	Polarity	Contoh Jenis Filer
SMAW	DCEP/DCRP	E 6010
GTAW	DCEN/DCSP	ER 70S

E 309 = Filler Metal untuk CS/SS

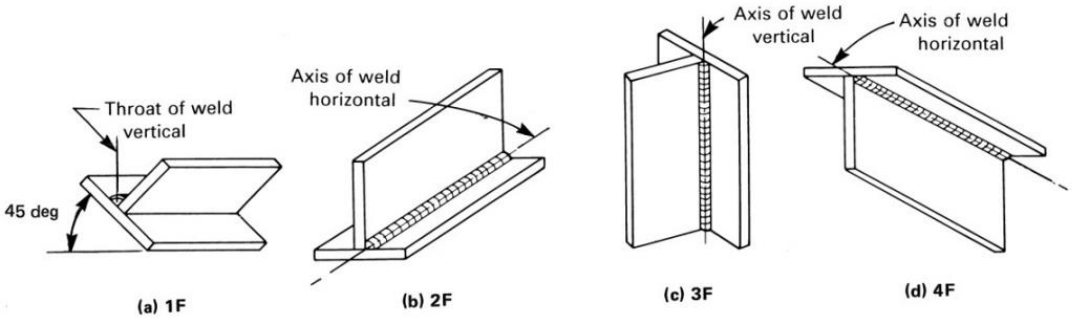
QW-461.3 GROOVE WELDS IN PLATE — TEST POSITIONS



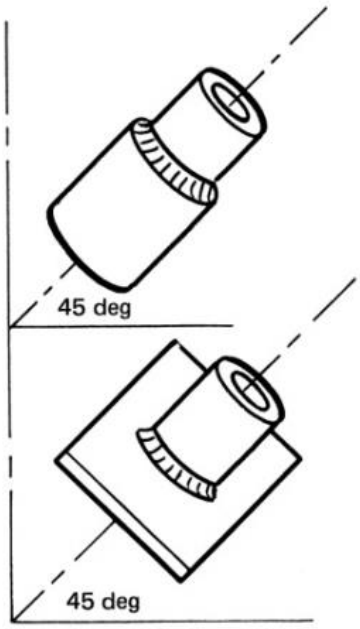
QW-461.4 GROOVE WELDS IN PIPE — TEST POSITIONS



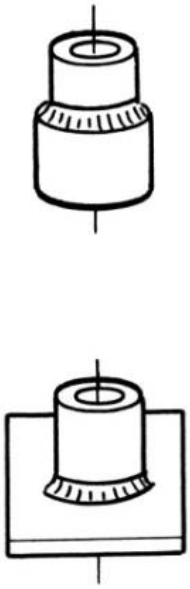
QW-461.5 FILLET WELDS IN PLATE — TEST POSITIONS



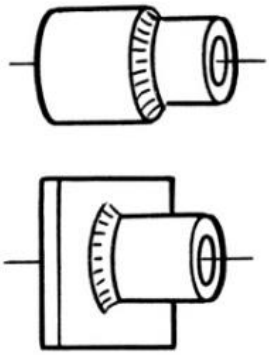
QW-461.6 FILLET WELDS IN PIPE — TEST POSITIONS



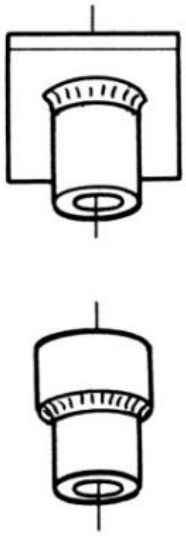
(a) 1F (Rotated)



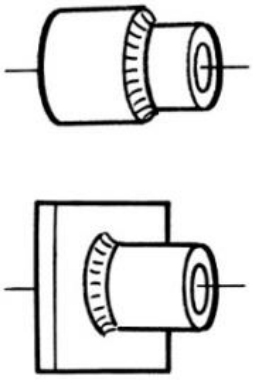
(b) 2F



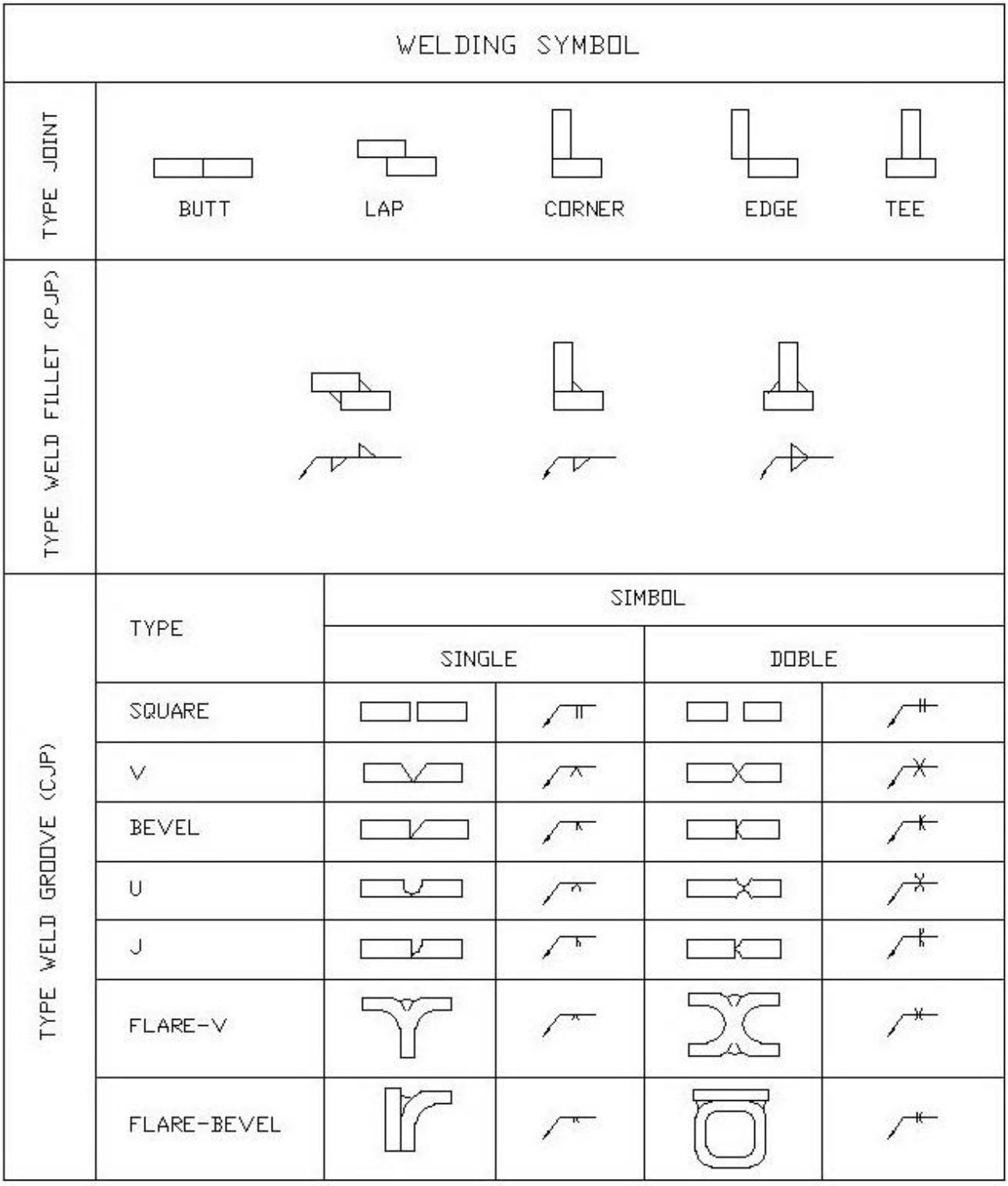
(c) 2FR (Rotated)



(d) 4F



(e) 5F



RADIOGRAPHY

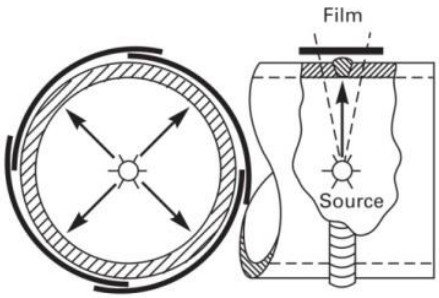
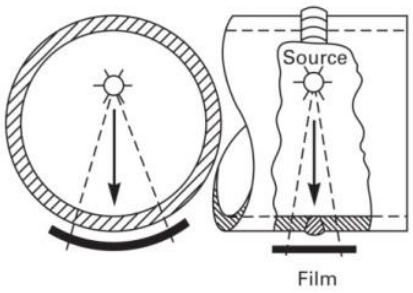
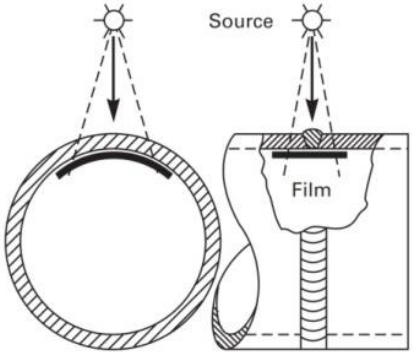
T-270 EXAMINATION

T-271 Radiographic Technique (Sketches showing suggested source, film, and IQI placements for pipe or tube welds are illustrated in Article 2, Nonmandatory Appendix A.)

1. Sigle Wall Teknik Single Wall View
2. Doble Wall Teknik Single Wall View
3. Doble Wall Teknik Doble Wall View

1. Singe Wall Technique_Singe View Imeg.
Contoh: A/B/C

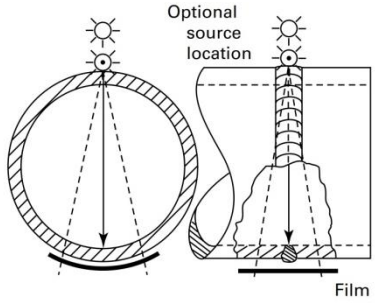
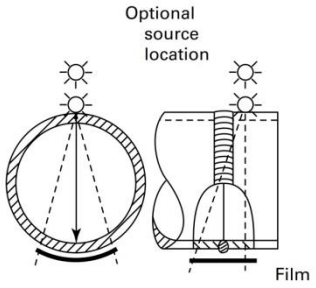
FIG. A-210-1 SINGLE-WALL RADIOGRAPHIC TECHNIQUES

O.D.	Exposure Technique	Radiograph Viewing	Source-Weld-Film Arrangement		IQI		Location Marker Placement
			End View	Side View	Selection	Placement	
Any	Single-Wall T-271.1	Single-Wall	 Exposure Arrangement — A		T-276 and Table T-276	Source Side T-277.1(a) Film Side T-277.1(b)	Either Side T-275.3 T-275.1(c)
Any	Single-Wall T-271.1	Single-Wall	 Exposure Arrangement — B		T-276 and Table T-276	Source Side T-277.1(a) Film Side T-277.1(b)	Film Side T-275.1 (b)(1)
Any	Single-Wall T-271.1	Single-Wall	 Exposure Arrangement — C		T-276 and Table T-276	Source Side T-277.1(a) Film Side T-277.1(b)	Source Side T-275.1 (a)(3)

2. Doble Wall Technique_Single View Imeg.

Contoh:D/E

FIG. A-210-2 DOUBLE-WALL RADIOGRAPHIC TECHNIQUES

O.D.	Exposure Technique	Radiograph Viewing	Source-Weld-Film Arrangement		IQI		Location Marker Placement
			End View	Side View	Selection	Placement	
Any	Double-Wall: T-271.2(a) at Least 3 Exposures 120 deg to Each Other for Complete Coverage	Single-Wall	 Exposure arrangement — D		T-276 and Table T-276	Source Side T-277.1(a) Film Side T-277.1(b)	Film Side T-275.1 (b)(1)
Any	Double-Wall: T-271.2(a) at least 3 Exposures 120 deg to Each Other for Complete Coverage	Single-Wall	 Exposure arrangement — E		T-276 and Table T-276	Source Side T-277.1(a) Film Side T-277.1(b)	Film Side T-275.1 (b)(1)

3. Doble Wall Technique_Doble View Imeq.

Contoh: F/G

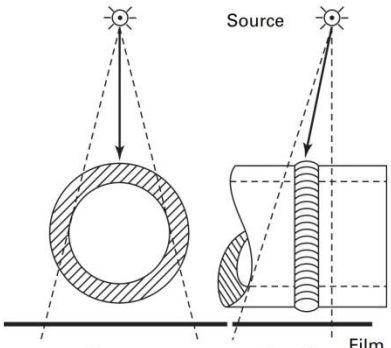
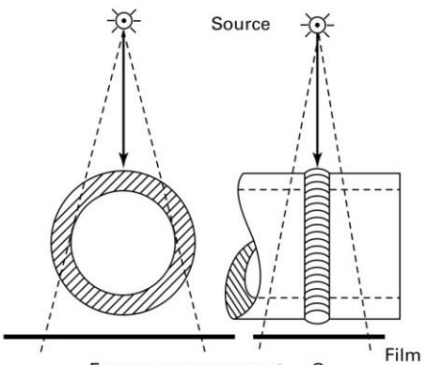
3½ in. (88 mm) or Less	Double-Wall T-271.2(b)(1) at Least 2 Exposures at 90 deg to Each Other for Complete Coverage	Double-Wall (Ellipse): Read Offset Source Side and Film Side Images	 Exposure arrangement — F	T-276 and Table T-276	Source Side T-277.1(a)	Either Side T-275.2

FIG. A-210-2 DOUBLE-WALL RADIOGRAPHIC TECHNIQUES (CONT'D)

O.D.	Exposure Technique	Radiograph Viewing	Source-Weld-Film Arrangement		IQI		Location Marker Placement
			End View	Side View	Selection	Placement	
3½ in. (88 mm) or Less	Double-Wall: T-271.2(b)(2) at Least 3 Exposures at 60 deg or 120 deg to Each Other for Complete Coverage	Double-Wall: Read Superimposed Source Side and Film Side Images	 Exposure arrangement — G		T-276 and Table T-276	Source Side T-277.1(a)	Either Side T-275.2